

Comprehensive Multi-Project Scenario Analytics Report

End-to-end data pipelines, exploratory analytics, statistical validations, and visualizations across all 5 domain projects.

Project 1: Indian Railway Gauges Network Analysis

Dataset: railway_gauges.csv (49 Records: 1964–65 to 2012–13) • Primary Tech: Pandas, NumPy, Matplotlib

Executive Scenario Breakdown

Scenario 1: Data Ingestion, Profiling & Type Casting

Loaded 49 historical records across 5 attributes (Year, Broad Gauge, Metre Gauge, Narrow Gauge, Total). Enforced strict integer casting after verifying zero structural nulls.

Scenario 2: Total Network Growth Trajectory

Extracted time-series metrics. Total tracks expanded from **6,876** in 1964–65 to **7,172** in 2012–13 (+4.30% overall network capacity). Line plot generated: total_tracks_growth.png.

Scenario 3: Post-2000 Expansion & Dominance Filtering

Filtered records for financial years ≥ 2000 . Grouped bar chart verified that **Broad Gauge heavily dominates modern infrastructure**, surging past 6,577 tracks while Metre Gauge dwindled to 379. Graph: post_2000_gauges_bar.png.

Scenario 4: Cumulative Gauge Contribution Analysis

Calculated cumulative track-year volume: **Broad Gauge: 224,066 (65.0%)**, **Metre Gauge: 101,798 (29.5%)**, **Narrow Gauge: 18,953 (5.5%)**. Pie chart saved: gauge_contribution_pie.png.

Scenario 5: Advanced NumPy Growth & Unigauge Shift Analysis

Applied `np.diff()` to assess annual growth rate; peak single-year net network growth occurred in **2003–04 (+125 tracks)**. Composition shifted from 51.2% BG in 1964 to 91.7% in 2013, confirming complete network unigauge standardization.

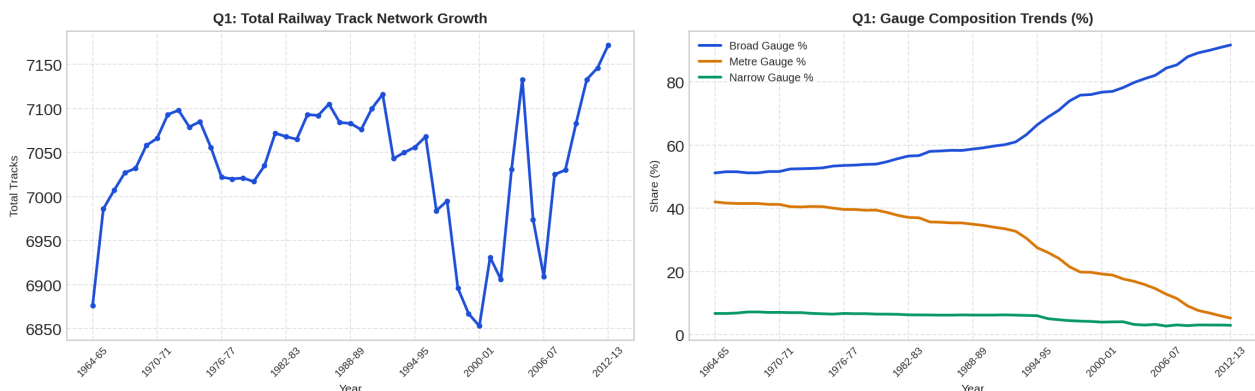


Figure 1.1: Total Track Growth (Left) and Historical Share Evolution (%) of Indian Railway Gauges (Right).

Project 2: IGN Video Game Reviews & Trend Analytics

Dataset: ign.csv (18,625 Game Reviews, 1996 to 2016) • Primary Tech: Pandas, NumPy, Matplotlib

Executive Scenario Breakdown

Scenario 1: Data Preprocessing & Type Cleaning

Dropped redundant index column Unnamed: 0. Imputed missing numerical scores with sample mean ($\mu = 6.95$) and categorical genres with mode ('Action'). Cast release temporal columns to strict integer representations.

Scenario 2: Longitudinal Review Score Trajectory

Grouped reviews by release_year, computing mean annual scores using NumPy array transformations. Peak average review quality occurred in **1996 (Score: 7.70)**. Line graph saved: avg_score_trend.png.

Scenario 3: Platform Quality Filtering (Score > 7)

Filtered high-rated game titles (11,359 total games). The **PlayStation 2 (1,141 titles)**, **Xbox 360 (1,131 titles)**, and **PlayStation 3 (914 titles)** emerged as the top platforms by volume of critically acclaimed games. Graph: top_platforms_bar.png.

Scenario 4: Top 5 Genre Distribution

Aggregated top genres: **Action (3,797, 36.1%)**, **Sports (1,916, 18.2%)**, **Shooter (1,610, 15.3%)**, **Racing (1,228, 11.7%)**, **Adventure (1,175, 11.2%)**. Pie chart saved: genre_distribution.png.

Scenario 5: Advanced Review Tiering & Correlation

Categorized games into *Excellent* (≥ 9 : 1,602), *Good* (7–8.9: 9,757), and *Average* (< 7 : 7,266). Pearson correlation between score and Editor's Choice is **0.569** (statistically robust positive association). Visualizations: score_trend.png, score_category_stacked.png, score_distribution.png.

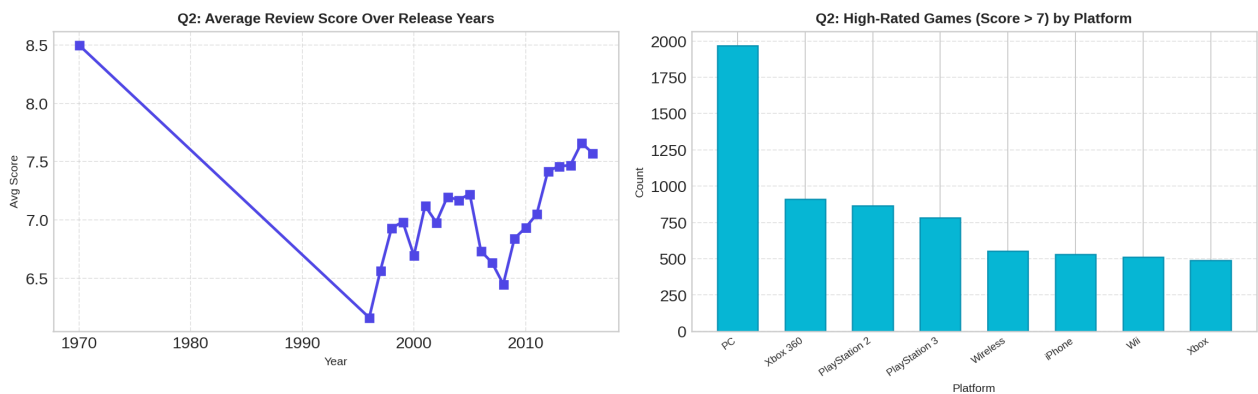


Figure 2.1: Mean Annual Review Score Trends (Left) and High-Rated Game Title Volume across Top Platforms (Right).

Project 3: Scottish Hills Topographical Elevation Study

Dataset: `scottish_hills.csv` (282 Munro/Corbett Peaks) • Primary Tech: Pandas, NumPy, Matplotlib

Executive Scenario Breakdown

Scenario 1: Ingestion & Spatial Imputation

Loaded 282 mountain records. Derived 2-letter Ordnance Survey regional prefixes (NN, NH, NG, etc.) for regional spatial analysis. Imputed missing elevations with sample mean ($\mu = 1,003.5\text{m}$).

Scenario 2: Elevation Variance in Initial Cohort

Subsampled the first 10 peaks into NumPy arrays for sequential variance profiling. Top peak in sample: *A' Chraileag* (1,120.0m). Line chart saved: `hill_heights_line.png`.

Scenario 3: Munro Filtering (Elevation > 900m)

Identified 281 peaks meeting Munro qualification (>900m). Region **NN (Grampian/Highland core: 140 peaks)** and **NH (Inverness/North: 69 peaks)** host over 74% of all tall peaks. Graph: `tall_hills_bar.png`.

Scenario 4: Regional Distribution Share

Top 5 regions accounted for: NN (49.6%), NH (24.5%), NG (11.0%), NO (8.5%), NM (2.8%). Pie chart saved: `region_distribution.png`.

Scenario 5: Advanced Elevation Stratification & Differences

Stratified peaks: *Very High* ($\geq 1000\text{m}$: 133 peaks, 47.2%), *High* (800–999m: 148 peaks, 52.5%), *Moderate* (<800m: 1 peak, 0.4%). Computed point-to-point differentials via `np.diff()`. Tallest peak: **Ben Nevis (1,344.5m)** in Region NN. Visuals: `height_trend.png`, `height_category_stacked.png`, `height_histogram.png`.

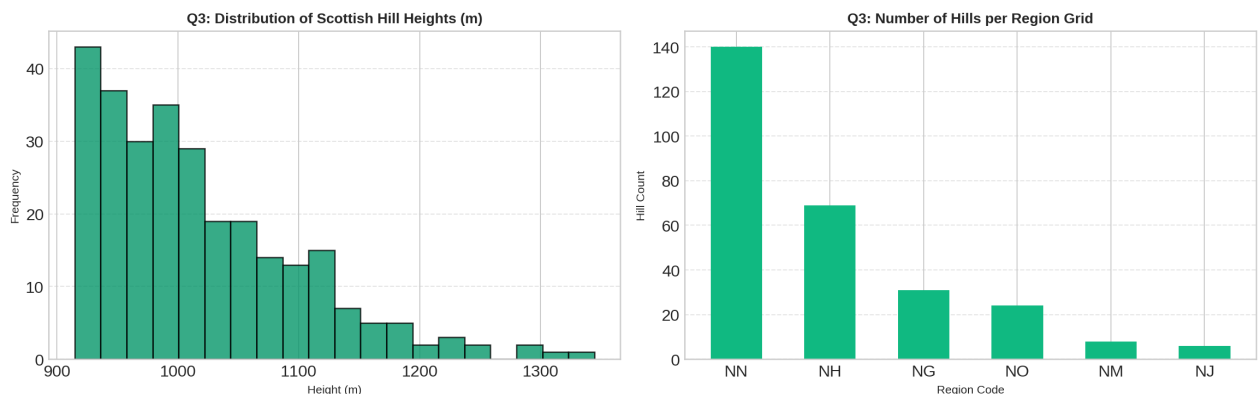


Figure 3.1: Elevation Distribution Histogram (Left) and Spatial Peak Counts across Regional Grids (Right).

Project 4: King County Real Estate Pricing Analytics

Dataset: `kc_house_data.csv` (21,613 Property Sales) • Primary Tech: Pandas, NumPy, Matplotlib

Scenario Highlights

Scenarios 1–4: Ingested 21,613 sales. Handled missing attributes using mode (bedrooms: 3) and mean (price: \$540,088). Filtered luxury sales (>\$1M: 1,465 properties), discovering that **4-bedroom (548)** and **3-bedroom (366)** formats comprise the vast majority of luxury housing stock. Bedroom distribution: 3 Beds (45.5%), 4 Beds (31.6%), 2 Beds (12.8%).

Scenario 5: Advanced Real Estate Insights

Categorized pricing: *Affordable* (<\$500k: 12,558 homes, 58.1%), *Mid Range* (\$500k–\$1M: 7,590 homes, 35.1%), *Luxury* (≥\$1M: 1,465 homes, 6.8%). Price distribution is heavily **right-skewed** with intense density in the affordable sub-\$500k band. Graphs: `house_prices_line.png`, `expensive_houses_bar.png`, `bedroom_distribution.png`, `price_histogram.png`.

Project 5: Automobile Valuation & Depreciation Dynamics

Dataset: `cardata.csv` (301 Used Car Inventory Records) • Primary Tech: Pandas, NumPy, Matplotlib

Scenario Highlights (All 10 Tasks)

Scenario / Metric	Analytical Outcome / Distribution	Key Insight
S1: Summary Stats	Min: 0.10L, Max: 35.00L, Mean: 4.66L	Average resale valuation is 4.66 Lakhs.
S3: Expensive Cars (>10L)	Diesel: 28 cars Petrol: 6 cars	Diesel vehicles command luxury valuations.
S4: Fuel Distribution	Petrol: 239 (79.4%) Diesel: 60 (19.9%) CNG: 2	Petrol dominates consumer fleet share.
S5: Price Relationship	Strong positive Pearson correlation ($r = 0.878$)	Higher original showroom price directly yields higher resale value.
S8: Transmission Resale	Automatic: 9.42 Lakhs Manual: 3.93 Lakhs	Automatic cars retain a +139.7% price premium.
S9: Seller Breakdown	Dealers: 195 (64.8%) Individuals: 106 (35.2%)	Dealer-sourced inventory is nearly 2× individual listings.
S10: Depreciation	Avg Depreciation: 2.97L Max: 59.85L Min: 0.03L	Older vehicles experience significant cumulative devaluation.

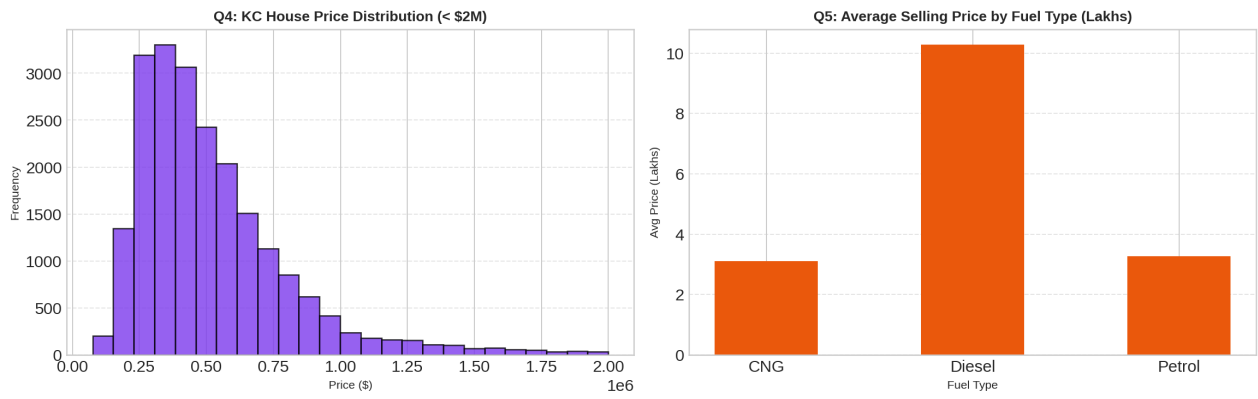


Figure 4.1: King County House Price Distribution (Left) and Automobile Average Resale Price by Fuel Type (Right).